1990 年数学(二) 真题解析

一、填空题

(1)【答案】 $y=\sqrt{3}x-1$.

【解】 当 $t=\dfrac{\pi}{6}$ 时,$x=\dfrac{3\sqrt{3}}{8}$,$y=\dfrac{1}{8}$;

$$\frac{dy}{dx}=\frac{dy/dt}{dx/dt}=\frac{3\sin^2 t\cos t}{-3\cos^2 t\sin t}=-\tan t,\quad \left.\frac{dy}{dx}\right|_{t=\frac{\pi}{6}}=-\frac{\sqrt{3}}{3},$$

则法线方程为 $y-\dfrac{1}{8}=\sqrt{3}\left(x-\dfrac{3\sqrt{3}}{8}\right)$,即 $y=\sqrt{3}x-1$.

(2)【答案】 $-\dfrac{1}{x^2}e^{\tan\frac{1}{x}}\left(\sec^2\dfrac{1}{x}\sin\dfrac{1}{x}+\cos\dfrac{1}{x}\right)$.

【解】 $y'=\left(e^{\tan\frac{1}{x}}\right)'\sin\dfrac{1}{x}+e^{\tan\frac{1}{x}}\left(\sin\dfrac{1}{x}\right)'$

$$=-\frac{1}{x^2}e^{\tan\frac{1}{x}}\sec^2\frac{1}{x}\sin\frac{1}{x}-\frac{1}{x^2}e^{\tan\frac{1}{x}}\cos\frac{1}{x}=-\frac{1}{x^2}e^{\tan\frac{1}{x}}\left(\sec^2\frac{1}{x}\sin\frac{1}{x}+\cos\frac{1}{x}\right).$$

(3)【答案】 $\dfrac{4}{15}$.

【解】 方法一

$$\int_0^1 x\sqrt{1-x}\,dx\xrightarrow{\sqrt{1-x}=t}\int_1^0(1-t^2)t\cdot(-2t)dt=2\int_0^1(t^2-t^4)dt$$

$$=2\left(\frac{1}{3}-\frac{1}{5}\right)=\frac{4}{15}.$$

方法二

$$\int_0^1 x\sqrt{1-x}\,dx=-\int_0^1[1-(1-x)]\sqrt{1-x}\,d(1-x)=\int_1^0(x-1)\sqrt{x}\,dx$$

$$=-\int_0^1(x^{\frac{3}{2}}-x^{\frac{1}{2}})dx=-\left(\frac{2}{5}-\frac{2}{3}\right)=\frac{4}{15}.$$

(4)【答案】 $>$.

【解】 当 $-2\leqslant x\leqslant -1$ 时,$-x^3>x^3$,从而 $e^{-x^3}>e^{x^3}$,于是 $\int_{-2}^{-1}e^{-x^3}dx>\int_{-2}^{-1}e^{x^3}dx$.

(5)【答案】 1.

【解】 $f[f(x)]=\begin{cases}1, & |f(x)|\leqslant 1,\\ 0, & |f(x)|>1.\end{cases}$

因为 $|f(x)|\leqslant 1$,所以 $f[f(x)]\equiv 1$.

二、选择题

(1)【答案】 (C).

【解】 $\dfrac{x^2}{x+1}-ax-b=\dfrac{(1-a)x^2-(a+b)x-b}{x+1}$,

由 $\lim\limits_{x\to\infty}\left(\dfrac{x^2}{x+1}-ax-b\right)=0$ 得 $\begin{cases}1-a=0,\\ a+b=0,\end{cases}$ 解得 $a=1,b=-1$,应选(C).

(2)【答案】 (B).

【解】 令$\int f(x)\mathrm{d}x = F(x)+C$,则有

$\mathrm{d}\left[\int f(x)\mathrm{d}x\right] = \mathrm{d}[F(x)+C] = [F(x)+C]'\mathrm{d}x = f(x)\mathrm{d}x$,应选(B).

(3)【答案】 (A).

【解】 由$f'(x) = [f(x)]^2$,有

$$f''(x) = 2f(x)f'(x) = 2f(x)[f(x)]^2 = 2![f(x)]^3,$$

$$f'''(x) = 2!\cdot 3[f(x)]^2 f'(x) = 3![f(x)]^4,$$

一般地,用数学归纳法,有$f^{(n)}(x) = n![f(x)]^{n+1}$. 应选(A).

(4)【答案】 (A).

【解】 $F'(x) = \left[\int_x^{\mathrm{e}^{-x}} f(t)\mathrm{d}t\right]' = f(\mathrm{e}^{-x})\cdot(\mathrm{e}^{-x})' - f(x)\cdot x'$

$= -\mathrm{e}^{-x}f(\mathrm{e}^{-x}) - f(x)$.

应选(A).

(5)【答案】 (B).

【解】 $\lim\limits_{x\to 0}F(x) = \lim\limits_{x\to 0}\dfrac{f(x)}{x} = \lim\limits_{x\to 0}\dfrac{f(x)-f(0)}{x} = f'(0)$,

因为$F(0) = f(0) = 0 \neq \lim\limits_{x\to 0}F(x)$,所以$x = 0$为$F(x)$的第一类间断点,应选(B).

三、【解】

(1) $\lim\limits_{x\to\infty}\left(\dfrac{x+a}{x-a}\right)^x = \lim\limits_{x\to\infty}\left[\left(1+\dfrac{2a}{x-a}\right)^{\frac{x-a}{2a}}\right]^{x\cdot\frac{2a}{x-a}} = \mathrm{e}^{2a\lim\limits_{x\to\infty}\frac{x}{x-a}} = \mathrm{e}^{2a}$,

由$\mathrm{e}^{2a} = 9$得$a = \ln 3$.

(2) **方法一**

由$2y-x = (x-y)\ln(x-y)$,得$2\mathrm{d}y - \mathrm{d}x = (\mathrm{d}x-\mathrm{d}y)\ln(x-y) + \mathrm{d}x - \mathrm{d}y$,解得

$$\mathrm{d}y = \frac{x}{2x-y}\mathrm{d}x.$$

方法二

$2y-x = (x-y)\ln(x-y)$两边求导,得$2y'-1 = (1-y')\ln(x-y)+1-y'$,

把$\ln(x-y) = \dfrac{2y-x}{x-y}$代入,解得$y' = \dfrac{x}{2x-y}$,故$\mathrm{d}y = \dfrac{x}{2x-y}\mathrm{d}x$.

(3) $y' = -\dfrac{2x}{(1+x^2)^2}$,

由$y'' = \dfrac{6x^2-2}{(1+x^2)^3} = 0$得$x = \dfrac{\sqrt{3}}{3}$,

当$0 < x < \dfrac{\sqrt{3}}{3}$时,$y'' < 0$;当$x > \dfrac{\sqrt{3}}{3}$时,$y'' > 0$,

又$y\Big|_{x=\frac{\sqrt{3}}{3}} = \dfrac{3}{4}$,故曲线$y = \dfrac{1}{1+x^2}$的拐点为$\left(\dfrac{\sqrt{3}}{3},\dfrac{3}{4}\right)$.

(4) $\int\dfrac{\ln x}{(1-x)^2}\mathrm{d}x = -\int\ln x\,\mathrm{d}\left(\dfrac{1}{x-1}\right) = -\dfrac{\ln x}{x-1} + \int\dfrac{1}{x(x-1)}\mathrm{d}x$

$$= \frac{\ln x}{1-x} + \int \left(\frac{1}{x-1} - \frac{1}{x} \right) \mathrm{d}x = \frac{\ln x}{1-x} + \ln \frac{|1-x|}{x} + C.$$

(5) $x \ln x \mathrm{d}y + (y - \ln x)\mathrm{d}x = 0$ 化为 $\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{1}{x \ln x} y = \frac{1}{x}$，解得

$$y = \left(\int \frac{1}{x} \mathrm{e}^{\int \frac{1}{x \ln x} \mathrm{d}x} \mathrm{d}x + C \right) \mathrm{e}^{-\int \frac{1}{x \ln x} \mathrm{d}x} = \left(\frac{1}{2} \ln^2 x + C \right) \cdot \frac{1}{\ln x},$$

由 $y(\mathrm{e}) = 1$ 得 $C = \frac{1}{2}$，故 $y = \frac{\ln^2 x + 1}{2 \ln x}$.

四、【解】 设切点为 $P(x, y)$，

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ 两边对 x 求导得 $\frac{2x}{a^2} + \frac{2y}{b^2} y' = 0$，解得 $y' = -\frac{b^2 x}{a^2 y}$，

设切线方程为 $Y - y = -\frac{b^2 x}{a^2 y}(X - x)$，

由 $Y = 0$ 得 $X = \frac{a^2 y^2}{b^2 x} + x = \frac{a^2}{x}\left(\frac{y^2}{b^2} + \frac{x^2}{a^2} \right) = \frac{a^2}{x}$，

由 $X = 0$ 得 $Y = \frac{b^2 x^2}{a^2 y} + y = \frac{b^2}{y}\left(\frac{x^2}{a^2} + \frac{y^2}{b^2} \right) = \frac{b^2}{y}$，

得所围成的面积为 $S = \frac{a^2 b^2}{2xy} - \frac{\pi ab}{4}$.

令 $F = \frac{a^2 b^2}{2xy} - \frac{\pi ab}{4} + \lambda \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} - 1 \right)$，

由 $\begin{cases} F'_x = -\frac{a^2 b^2}{2x^2 y} + \frac{2\lambda}{a^2} x = 0, \\ F'_y = -\frac{a^2 b^2}{2xy^2} + \frac{2\lambda y}{b^2} = 0, \text{得 } y = \frac{b}{a} x, \text{从而有 } x = \frac{a}{\sqrt{2}}, y = \frac{b}{\sqrt{2}}, \\ F'_\lambda = \frac{x^2}{a^2} + \frac{y^2}{b^2} - 1 = 0, \end{cases}$

得所求的切点坐标为 $\left(\frac{a}{\sqrt{2}}, \frac{b}{\sqrt{2}} \right)$.

五、【证明】 令 $f(x) = \arctan x + \frac{1}{x} - \frac{\pi}{2}$，

由 $f'(x) = \frac{1}{1+x^2} - \frac{1}{x^2} < 0 (x > 0)$ 得 $f(x)$ 在 $(0, +\infty)$ 内单调减少，

因为 $\lim\limits_{x \to +\infty} f(x) = 0$，所以当 $x > 0$ 时，$f(x) > 0$，即 $\arctan x + \frac{1}{x} > \frac{\pi}{2}$.

六、【解】 $f\left(\frac{1}{x} \right) = \int_1^{\frac{1}{x}} \frac{\ln t}{1+t} \mathrm{d}t \xlongequal{\frac{1}{t} = u} \int_1^x \frac{-\ln u}{1 + \frac{1}{u}} \left(-\frac{1}{u^2} \right) \mathrm{d}u = \int_1^x \frac{\ln u}{u(u+1)} \mathrm{d}u$

$$= \int_1^x \frac{\ln t}{t(t+1)} \mathrm{d}t = \int_1^x \frac{\ln t}{t} \mathrm{d}t - \int_1^x \frac{\ln t}{t+1} \mathrm{d}t, \text{则}$$

$$f(x) + f\left(\frac{1}{x} \right) = \int_1^x \frac{\ln t}{t} \mathrm{d}t = \frac{1}{2} \ln^2 t \Big|_1^x = \frac{1}{2} \ln^2 x.$$

七、【解】 设切点坐标为$(a,\sqrt{a-2})$，

由题意得$\dfrac{1}{2\sqrt{a-2}}=\dfrac{\sqrt{a-2}}{a-1}$，解得$a=3$，则切点为$(3,1)$，

切线方程为$y=\dfrac{1}{2}(x-1)$，

所求的体积为

$$V=\pi\int_1^3\frac{1}{4}(x-1)^2\mathrm{d}x-\pi\int_2^3(\sqrt{x-2})^2\mathrm{d}x$$

$$=\frac{\pi}{12}(x-1)^3\Big|_1^3-\frac{\pi}{2}(x-2)^2\Big|_2^3=\frac{\pi}{6}.$$

八、【解】 特征方程为$\lambda^2+4\lambda+4=0$，特征根为$\lambda_1=\lambda_2=-2$，

$y''+4y'+4y=0$的通解为$y=(C_1+C_2x)\mathrm{e}^{-2x}$($C_1,C_2$为任意常数)；

(1) 当$a\neq-2$时，$y''+4y'+4y=\mathrm{e}^{ax}$的特解为$y_0(x)=\dfrac{1}{(a+2)^2}\mathrm{e}^{ax}$，

原方程的通解为$y=(C_1+C_2x)\mathrm{e}^{-2x}+\dfrac{1}{(a+2)^2}\mathrm{e}^{ax}$　(C_1,C_2为任意常数)；

(2) 当$a=-2$时，令特解为$y_0(x)=bx^2\mathrm{e}^{-2x}$，代入得$b=\dfrac{1}{2}$，

原方程的通解为$y=(C_1+C_2x)\mathrm{e}^{-2x}+\dfrac{1}{2}x^2\mathrm{e}^{-2x}$　(C_1,C_2为任意常数).